



Festival Queue Merges

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Difficulty Level : Easy

Festival Queue Merges

A city festival has opened K entry gates. Each gate has its own queue of visitors. During the day, the organisers keep re-routing people between gates to balance traffic.

You are given Q operations that modify queue sizes. After each **QUERY** operation, you must report the busiest gate.

If multiple gates have the same maximum queue size, choose the **smallest gate index**.

Task: Process all operations and print the answer for every **QUERY**.

Input

The first line contains two integers K and Q ($1 \leq K, Q \leq 2 \cdot 10^5$) - the number of gates and the number of operations.

Each of the next Q lines is one of the following operations:

- **ADD** $g\ x$: add x people to gate g .
- **SERVE** $g\ x$: remove up to x people from gate g . (Queue size cannot go below 0.)
- **MOVE** $a\ b\ x$: move up to x people from gate a to gate b .
- **MERGE** $a\ b$: append all people from gate b to gate a , then gate b becomes empty.
- **QUERY** : ask for the busiest gate.

All indices satisfy $1 \leq g, a, b \leq K$, and for **MOVE/MERGE**, $a \neq b$. All amounts satisfy $1 \leq x \leq 10^9$.

Output

For each **QUERY**, print one line with two integers:

- the gate index with the maximum current queue size,
- that queue size.

Break ties by choosing the smallest gate index.

Constraints

- $1 \leq K, Q \leq 2 \cdot 10^5$

- $1 \leq x \leq 10^9$
- Time limit: **2 seconds**
- Memory limit: **256 MB**

Sample Input 1

```
4 11
ADD 1 5
ADD 2 3
QUERY
MOVE 1 2 4
QUERY
SERVE 2 2
ADD 3 9
MERGE 2 3
QUERY
ADD 4 14
QUERY
```

Sample Output 1

```
1 5
2 7
2 14
2 14
```

Explanation

- After the first two operations: queues are [5, 3, 0, 0] -> QUERY gives 1 5.
- MOVE 1 2 4 makes queues [1, 7, 0, 0] -> QUERY gives 2 7.
- SERVE 2 2, ADD 3 9, MERGE 2 3 makes queues [1, 14, 0, 0] -> QUERY gives 2 14.
- ADD 4 14 makes [1, 14, 0, 14]; tie between gates 2 and 4, pick smaller index -> 2 14.